Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/31/2008
PI Name:	Fornace, Albert M.D.		
Project Title:	Mouse models approach for intestinal tumorigenesis estimates by space radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	http://www9.georgetown.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
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No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start/end dates changed to 4/1/07-3/31/11, per update from P. Denkins/JSC; previously had 12/1/2006-11/30/2010 (9/2007)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Ionizing radiation is a known risk factor for colorectal (CR) cancer. Considering the high spontaneous incidence of gastrointestinal (GI) cancer in the US and even higher incidence of pre-malignant lesions, such as colonic polyps, an even modest increase by radiation exposure could have a significant effect on health risk estimates for manned space flight. During space travel, astronauts are exposed to high levels of radiation, which increases their risk for acute short-term deleterious radiation effects and for long term deleterious effects like cancer. Long duration space missions, like those at the International Space Station, lunar missions, or a mission to Mars, increase astronauts' exposure to space radiation as well as the risks associated with such exposure. This project is important because there are no good estimates for intestinal, particularly CR, cancer by space radiation. Considering the increased CR cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors and to investigate modulation of specific signaling pathways as a model for cancer prevention. We are employing an animal model, APC+/min, that has been successfully used to demonstrate radiation-induced intestinal cancer. The APC+/min model has inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. The strategy is to compare results for space radiation, protons and HZE (Iron-56), on intestinal tumor development to that of low-LET (linear energy transfer) radiation (gamma-rays). Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for humans exposed to space radiation. Molecular events leading to intestinal tumors have been characterized. We will monitor and compare molecular changes in three types of radiation-induced tumors. The specific aims for this proposal are: 1. Quantitatively compare HZE ion or high-energy proton irradiation to gamma-ray effects on intestinal tumor incidence and grade using the Apc+/min tumor model. 2. Characterize and compare inactivation of wt APC locus on chromosome 18B1 after gamma-ray vs. HZE ion or high-energy proton irradiation with the Apc+/min tumor model. 3. Determine intestinal tumor incidence in the Apc+/min; Ppm1d-/- mouse model and mechanisms for reactivation of the Apc+/min tumor phenotype by radiation. 4. Determine molecular signatures by comparative genomic hybridization (CGH) and expression profiling in spontaneous intestinal tumors compared to those induced by HZE ions, high-energy protons, and gamma-irradiation. 5. Quantitatively compare HZE ion or high-energy proton irradiation vs. gamma-ray effects on incidence of other tumors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and the incidence of CR cancer in 2002 was 53.6 per 100000 with a death rate of 20.1 per 100000 or about 57,000 deaths per year. Considering the high frequency of CR tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and CR carcinogenesis.
Task Progress:	The overall plan is to apply the well-characterized APC +/min mouse model of intestinal tumorigenesis for development of risks estimates for various radiation exposures during manned space flight. To undertake a comparative study of intestinal tumor risk in mice following exposure to different types of radiation, preliminary experiments are underway to determine the relative toxicity of Fe-56 particles and protons to gamma-rays, so that we can standardize with doses equitoxic to 2 and 5 Gy of gamma-radiation in C57Bl/6J mice; these gamma-ray doses have been shown to be effective in increasing intestinal tumorigenesis. The rationale for this preliminary study is that acute toxicity for the planned radiation sources is not available. We have already carried out two runs in the first year of this project at the NSRL at Brookhaven National Laboratory. Results to date indicate that the relative biological effectiveness (RBE) for Fe-56 particles compared to gamma-rays is 1.29. This RBE is surprisingly low and is an important parameter that will allow us to plan long-term experiments. Irradiated mice are being observed long-term, and specific tissues are also being provided to various NASA investigators to complement their own projects. In our second run (Sept 2007) we also irradiated APC+/min mice with iron particles and protons. A total of 20 mice were irradiated and are being monitored for intestinal tumorigenesis. While not statistically significant, it is interesting that a few irradiated mice developed high intestinal tumor load within the first 100 days after irradiation, while this was not seen in unirradiated mice. At the end of the observation period (100-150 days), mice will be euthanized for intestinal tumor counting, as well as for genetic and molecular pathway analysis of individual tumors.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Datta K, Doiron K, Fornace A Jr. "Low- and high-LET radiation induced chronic stress responses in mice: Implications for carcinogenesis." To be presented at the 99th Annual meeting of the American Association for Cancer Research, San Diego, CA, April 12- 16, 2008. American Association for Cancer Research (AACR) annual meeting, April 12- 16, 2008. , Apr-2008